Homework09

Task 1 (1 points):

Open your GitHub repository in Codespace.

1. In the terminal run the command: `cd HW/Homework09/University/`
2. In the terminal run the command: `cp ../../../Labs/Lab03/University/.env .env`
3. In the terminal run the command: `cp ../../Homework08/Mystery/Database.py Database.py`

Task 2 (2 points):

At the top of the **Database.py** file in **HW/Homework09/University/**, also import: `sql from psycopg2`
Add the following methods to the **Database** class:

1. `def reset_tables(self) -> None:`
   Call the **drop_tables()**, **create_tables()**, and **load_data()** methods in that order.
2. `def factory_reset(self) -> None:`
   Define this method using the factory_reset() function from reset.py. At the beginning of the try block, check if self.connection is not None. If it is, close the connection.

Task 3 (2 points):

Modify the **connect()** method such that is has the following function header:

```
def connect(self, username: str = None, password: str = None) -> None:
```

1. For the **user** variable assignment, provide a ternary expression that assigns `os.getenv("DB_USER")` to **user** if `username` is None. Otherwise it assigns `username` to **user**. (This should be a one-liner.)
2. For the **password** variable assignment, provide a ternary expression that assigns `os.getenv("DB_PASSWORD")` to **password** if `password` is None. Otherwise it assigns `password` to **password**. (This should be a one-liner.)

Task 4 (5 points):

Create file called **hw09.py** and import Database from Database as DB.

1. Instantiate a DB object called **db**. Then connect to the database calling the **connect()** method.
2. Execute the following queries:
   a. Create a role called cs_chair with a login password 'chair123'.
   b. Create a view called cs_instructors that contains all Comp. Sci. instructors.
   c. Grant all privileges to cs_chair on the cs_instructors view.
   d. Grant read privileges to cs_chair on the student relation.
3. Close the **db** connection.
4. Instantiate a DB object called **db_user**. Then connect to the database calling the **connect()** method with 'cs_chair' and 'chair123' as arguments.
5. Execute a query that shows all cs_instructors.
6. Execute a query that shows all students.
7. Close the **db_user** connection and connect to **db**.
8. Execute the following quieries:
   a. Drop the cs_instructors view.
   b. Revoke read privileges from cs_chair on the student relation.
   c. Drop the cs_chair role.
9. Close the db connection.
10. Run **hw09.py**.

Push your work to GitHub using the following commands:

```
git pull
git add .
git commit -m "homework09 upload"
git push
```